SoK: Enhancing Cryptographic Collaborative Learning with Differential Privacy

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TABLE I: Categorization of CPCL Papers by Learning Paradigm, DP Noise and security properties. Perturbations are Grad(ient), Out(put). Noise Mechanisms include Dist(ributed) Lap(lace), Disc(rete) Gauss(ian), Poisson-Bin(omial). ✓/× denote existing/missing features, — gaps , and ⊛ potential enhancements. Privacy Unit is R(ecord) or U(ser) level. DP Analysis includes Mom(ents) Acc(ountant) and C(entral) L(imit) T(heorem) [1]. We indicate Malicious $\mathcal{C}/\mathcal{S}$, and Collusion thresholds.

Learning Paradigm	Type (Alg. 2)	Sampling (Fig. 3)	Noise Perturb (Alg. 1)	Mechanism (Tab. II)	Sampler	Gradient Secrecy $\mathcal{C}$	Gradient Secrecy $\mathcal{S}$	Model Secrecy $\mathcal{C}$	Model Secrecy $\mathcal{S}$	Oblivious Noise	Papers	Privacy Unit	DP Analysis	Cryptographic Techniques	Malicious	Collusion
FL	PNoise	Local	Grad.	Binomial	$\mathcal{C}$	×	⊛	×	⊛	⊛	[2]	U	(ϵ, δ)	Masking	×	×
				Disc. Gauss.	$\mathcal{C}$	×	⊛	×	⊛	⊛	[3]	U	RDP	Masking	×	×
					$\mathcal{C}$	×	⊛	×	⊛	⊛	[4]	R	RDP	Masking	×	×
					$\mathcal{C}$	×	⊛	×	⊛	⊛	[5]	R	RDP	LWE	$\mathcal{C}, \mathcal{S}$	$n/2$
				Skellam	$\mathcal{C}$	×	⊛	×	⊛	⊛	[6]	U	RDP	Masking	×	×
					$\mathcal{C}$	×	⊛	×	⊛	⊛	[7]	R	RDP	Masking	×	×
				Poisson-Bin.	$\mathcal{C}$	×	⊛	×	⊛	⊛	[8]	U	RDP	Masking	×	×
				Gaussian	$\mathcal{C}$	×	⊛	×	⊛	⊛	[9]	R	Mom. Acc.	(A)HE	×	$n-1$
					$\mathcal{C}$	×	✓	×	✓	⊛	[10]	U	Mom. Acc.	(F)HE	×	$n-1$
					$\mathcal{C}$	×	⊛	×	⊛	⊛	[11]	R	RDP	(A)HE	×	$n-1$
					$\mathcal{S}$	×	⊛	×	⊛	⊛	[12]	R	CLT	SS	$\mathcal{C}$	$n/2, m-1$
					$\mathcal{C}$	×	⊛	×	⊛	⊛	[13]	U	RDP	Masking	$\mathcal{C}, \mathcal{S}$	$n-1$
			Out.	Dist. Lap.	$\mathcal{C}$	✓	✓	×	⊛	✓	[14]	R	ϵ	Masking	×	$n-1$
					$\mathcal{C}$	✓	✓	×	⊛	✓	[15]	R	ϵ	A(HE)	×	$n-1$
					$\mathcal{C}$	✓	✓	×	⊛	✓	[16]	R	ϵ	A(HE)	$\mathcal{S}$	$n-1$
	CNoise	Distributed	Grad.	Gaussian	$\mathcal{S}$	×	⊛	×	⊛	✓	[17]	R	Mom. Acc.	SS + GC	×	×
					$\mathcal{S}$	×	⊛	×	⊛	✓	[18]	R	zCDP	SS + GC	$\mathcal{S}$	×
					$\mathcal{S}$	×	⊛	×	⊛	✓	[19]	R	(ϵ, δ)	SS + GC	×	×
			Out.	Laplace	$\mathcal{S}$	✓	✓	×	⊛	✓	[18]	R	ϵ	SS + GC	$\mathcal{S}$	×
		Centralized	Grad.	—	—	×	⊛	×	⊛	✓	—	—	—	—	—	—
			Out.	Laplace	$\mathcal{S}$	✓	✓	×	⊛	✓	[20]	R	ϵ	(A)HE	$\mathcal{C}, \mathcal{S}$	$m-1$
OL	PNoise	Local	Grad.	Binomial	$\mathcal{S}$	✓	✓	✓	✓	⊛	[21]	R	RDP	SS	×	$m/2$
				Disc. Gauss.	$\mathcal{S}$	✓	✓	✓	✓	⊛	[22]	R	Mom. Acc.	SS	×	$m-1$
			Out.	—	—	✓	✓	✓	✓	⊛	—	—	—	—	—	—
	CNoise	Distributed	Grad.	—	—	✓	✓	✓	✓	✓	—	—	—	—	—	—
			Out.	Dist. Lap.	$\mathcal{S}$	✓	✓	✓	✓	✓	[23]	R	ϵ	SS	$\mathcal{S}$	$m-1$
		Centralized	—	—	—	✓	✓	✓	✓	✓	—	—	—	—	—	—